Singular Value Decomposition (SVD) and Principal Component Analysis

Singular Value Decomposition (SVD) represents one of the most powerful theorems in applied linear algebra. Given any real m-by-n matrix A, SVD decomposes it into the product of three canonical matrices: A = U * Sigma * V^T, where U is an m-by-m orthogonal matrix of left singular vectors, Sigma is an m-by-n diagonal matrix containing non-negative singular values sorted in descending order, and V^T is the transpose of an n-by-n orthogonal matrix of right singular vectors.

In data science and dimensionality reduction, SVD forms the mathematical foundation of Principal Component Analysis (PCA). The top singular values quantify the variance of data points projected along the corresponding singular vectors. By truncating Sigma to retain only the top k singular values, we obtain the optimal low-rank approximation of matrix A under the Frobenius norm (Eckart-Young-Mirsky theorem), filtering noise while compressing multidimensional embeddings.